COMPREHENSIVE LOGARITHMS

Complete Study Notes: Basic Fundamentals to Advanced Concepts & Problem Solving

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1. Fundamentals of Logarithms

A logarithm is the inverse operation to exponentiation. It answers the question: *"To what power must we raise a given base to obtain a certain number?"*

If $b^y = x$, then $\log_b(x) = y$ (where $b > 0, b \neq 1$, and $x > 0$)

Mandatory Domain Constraints:

- **Argument (x):** Must be strictly positive ($x > 0$). Logarithm of zero or negative numbers is undefined in real numbers.
- **Base (b):** Must be positive and not equal to 1 ($b > 0, b \neq 1$).

Standard Types of Logarithms

- **Common Logarithm:** Base 10, written as $\log(x)$ or $\log_{10}(x)$.
- **Natural Logarithm:** Base e ($e \approx 2.71828$), written as $\ln(x)$ or $\log_e(x)$.

2. Fundamental Properties & Rules

Property Name	Mathematical Statement	Conditions / Description
Product Rule	$\log_b(m \times n) = \log_b(m) + \log_b(n)$	Converts multiplication inside log to addition outside.
Quotient Rule	$\log_b(m / n) = \log_b(m) - \log_b(n)$	Converts division inside log to subtraction outside.
Power Rule	$\log_b(m^k) = k \cdot \log_b(m)$	Exponent on argument moves to the front as multiplier.
Base Power Rule	$\log_{b^k}(m) = (1 / k) \cdot \log_b(m)$	Exponent on base moves to front as a reciprocal factor.
Change of Base	$\log_b(a) = \log_c(a) / \log_c(b)$	Allows changing base to any convenient positive base c.
Reciprocal Base	$\log_b(a) = 1 / \log_a(b)$	Swapping base and argument inverts the logarithmic value.
Identity Properties	$\log_b(b) = 1$ and $\log_b(1) = 0$	Logarithm of base itself is 1; logarithm of 1 is always 0.
		Powers and logarithms with matching bases cancel out.

Property Name	Mathematical Statement	Conditions / Description
Exponentiation Inversion	$b^{\log_b(x)} = x$ and $a^{\log_b(c)} = c^{\log_b(a)}$	

3. Advanced Logarithmic Concepts

A. Characteristic and Mantissa

Every common logarithm ($\log_{10}x$) consists of an integer part and a fractional part:

$$\log_{10}(x) = \text{Characteristic} + \text{Mantissa}$$

- **Characteristic:** The integer part. It can be positive, zero, or negative.
  - If $N > 1$, Characteristic = (Number of digits before decimal) − 1.
  - If $0 < N < 1$, Characteristic = − (Number of zeroes immediately after decimal + 1).
- **Mantissa:** The non-negative decimal fraction ($0 \le \text{Mantissa} < 1$). Found using logarithmic tables.

B. Logarithmic Inequalities

When solving logarithmic inequalities, the inequality sign depends on the value of the base b:

- **When base $b > 1$ (Increasing function):**

$$\log_b(f(x)) > \log_b(g(x)) \Rightarrow f(x) > g(x) > 0$$

- **When base $0 < b < 1$ (Decreasing function):**

$$\log_b(f(x)) > \log_b(g(x)) \Rightarrow 0 < f(x) < g(x)$$ *(Inequality direction flips!)*

4. Practice Problems with Step-by-Step Solutions

Level 1: Basic Level Problems

Problem 1 (Basic Evaluation): Find the value of $\log_2(32) + \log_3(1/27) - \log_5(\sqrt{5})$.

1. Evaluate $\log_2(32)$: Since $32 = 2^5$, $\log_2(2^5) = 5$.
2. Evaluate $\log_3(1/27)$: Since $1/27 = 3^{-3}$, $\log_3(3^{-3}) = -3$.
3. Evaluate $\log_5(\sqrt{5})$: Since $\sqrt{5} = 5^{1/2}$, $\log_5(5^{1/2}) = 1/2$.
4. Combine: $5 + (-3) - (1/2) = 2 - 0.5 = 1.5$.

Answer: 3/2 or 1.5

Problem 2 (Solving Basic Equation): Solve for x: $\log_{10}(x + 3) + \log_{10}(x - 3) = 1$.

1. Apply Product Rule: $\log_{10}[(x + 3)(x - 3)] = 1 \Rightarrow \log_{10}(x^2 - 9) = 1$.
2. Convert to exponential form: $x^2 - 9 = 10^1 = 10$.
3. Solve equation: $x^2 = 19 \Rightarrow x = \pm\sqrt{19}$.
4. Domain check: For $\log_{10}(x - 3)$ to be defined, $x - 3 > 0 \Rightarrow x > 3$. Thus, reject $x = -\sqrt{19}$.

Answer: $x = \sqrt{19}$

Level 2: Intermediate Level Problems

Problem 3 (Change of Base): Prove that $1 / \log_2(N) + 1 / \log_3(N) + 1 / \log_4(N) + \ldots + 1 / \log_{10}(N) = \log_N(10!)$.

1. Use reciprocal base rule $1 / \log_a(b) = \log_b(a)$ for each term:

   LHS = $\log_N(2) + \log_N(3) + \log_N(4) + \ldots + \log_N(10)$.
2. Apply Product Rule: $\log_N(2 \times 3 \times 4 \times \ldots \times 10)$.
3. Note that $1 \times 2 \times 3 \times \ldots \times 10 = 10!$.
4. LHS = $\log_N(10!) = RHS$. (Hence proved)

Result: Proved

Problem 4 (System of Equations): Solve for x and y: $\log_2(x) + \log_4(y) = 3$ and $\log_4(x) + \log_2(y) = 4$.

1. Convert base 4 terms using $\log_4(a) = (1/2)\log_2(a)$:

   Eq 1: $\log_2(x) + (1/2)\log_2(y) = 3 \Rightarrow 2\log_2(x) + \log_2(y) = 6$

   Eq 2: $(1/2)\log_2(x) + \log_2(y) = 4 \Rightarrow \log_2(x) + 2\log_2(y) = 8$
2. Let $a = \log_2(x)$ and $b = \log_2(y)$:

   $2a + b = 6$ and $a + 2b = 8$.
3. Subtracting equations: $a - b = -2 \Rightarrow b = a + 2$. Substitute: $2a + (a + 2) = 6 \Rightarrow 3a = 4 \Rightarrow a = 4/3$, hence $b = 10/3$.
4. Solve for x, y: $x = 2^{4/3} = 2\sqrt[3]{2}$, $y = 2^{10/3} = 8\sqrt[3]{2}$.

Answer: $x = 2^{4/3}$, $y = 2^{10/3}$

Level 3: Advanced Level Problems

Problem 5 (Logarithmic Inequality): Solve for x: $\log_{0.5}(x^2 - 5x + 6) \geq -1$.

1. **Domain Condition:** Argument must be positive: $x^2 - 5x + 6 > 0 \Rightarrow (x - 2)(x - 3) > 0 \Rightarrow x \in (-\infty, 2) \cup (3, \infty)$.
2. **Inequality Resolution:** Base $0.5 < 1$, so flipping inequality sign is required:

   $x^2 - 5x + 6 \leq (0.5)^{-1} \Rightarrow x^2 - 5x + 6 \leq 2$.

   $x^2 - 5x + 4 \leq 0 \Rightarrow (x - 1)(x - 4) \leq 0 \Rightarrow x \in [1, 4]$.
3. **Intersection:** Intersect Domain $((-\infty, 2) \cup (3, \infty))$ with Inequality Solution $[1, 4]$:

   $x \in [1, 2) \cup (3, 4]$.

Answer: x ∈ [1, 2) ∪ (3, 4]

Problem 6 (Advanced Functional Relation): If $a^2 + b^2 = 7ab$, prove that $\log[(a + b) / 3] = (1/2)[\log(a) + \log(b)]$.

1. Start with given relation: $a^2 + b^2 = 7ab$.
2. Add $2ab$ to both sides to complete the square:

   $a^2 + 2ab + b^2 = 9ab \Rightarrow (a + b)^2 = 9ab$.
3. Divide both sides by 9: $((a + b) / 3)^2 = ab$.
4. Take logarithm on both sides: $\log[((a + b) / 3)^2] = \log(ab)$.
5. Apply Power and Product rules: $2 \cdot \log[(a + b) / 3] = \log(a) + \log(b)$.
6. Divide by 2: $\log[(a + b) / 3] = (1/2)[\log(a) + \log(b)]$.

Result: Proved